

Brain Tumor Detection Project - Interview Preparation Report

Repository: yashwanth252005/Major_project

1) Project Overview

- Problem: MRI tumor prediction systems are often black-box; users need explainability.
- Solution: A web app that predicts brain tumor class and returns three explainability outputs (Grad-CAM, Integrated Gradients, SHAP).
- Main objective: combine model inference + interpretable visual outputs + scan history + PDF reporting.

2) Architecture (actual flow)

- User -> React frontend -> Django REST API -> Brain_Tumor_AI predictor -> SQLite persistence -> frontend visualization.
- Frontend routes: / (Dashboard), /history, /history/:id.
- Backend APIs: GET /api/health/, POST /api/predict/, GET /api/history/, GET/DELETE /api/history//.

3) Tech Stack (confirmed from code)

- Frontend: React 18, Vite, Tailwind CSS, Axios, React Router, Framer Motion, Lucide React, html2canvas, jsPDF.
- Backend: Django 4.2, Django REST Framework, django-cors-headers.
- ML/XAI: PyTorch, torchvision (DenseNet121), grad-cam, SHAP, Captum (Integrated Gradients), OpenCV, NumPy, Matplotlib, Pillow.
- Database: SQLite.

4) Core Modules

- Brain_Tumor_AI/predict.py: BrainTumorPredictor orchestrates preprocessing, prediction, and XAI generation.
- backend/api/inference.py: lazy singleton model loading with thread lock and path/cwd bridging.
- backend/api/views.py: upload validation, inference call, DB save, API responses.
- frontend/src/pages/Dashboard.jsx: upload + analyze workflow.
- frontend/src/pages/History.jsx and ScanDetail.jsx: history display, detail view, delete, export PDF.

5) Database (Scan table)

- Fields: id(UUID), predicted_class, confidence, processing_time, summary, original_image_b64, gradcam_b64, shap_b64, integrated_gradients_b64, created_at.
- Operations: Create on predict, list history, fetch detail, delete by UUID.

6) Security and Limitations (honest)

- No authentication/authorization; all API routes are AllowAny.
- ALLOWED_HOSTS = ["*"] and extension-only upload validation (no MIME/deep file checks).
- Error responses can leak exception details in predict endpoint.
- Base64 image storage in SQLite is simple but not scalable.
- No full automated test suite found in repository.

7) 30-second Interview Pitch

- "This is a brain MRI analysis project where we upload an MRI image, run a DenseNet-based model, and show not just the prediction but also three explainability maps: Grad-CAM, SHAP, and Integrated Gradients. We built a React frontend with a Django API backend, saved each scan in SQLite history, and added a report view with PDF export so results are easy to review."

8) Contribution Guidance

- Exact individual contribution cannot be fully confirmed from this shallow local git snapshot.
- Use your own GitHub username, commit/PR links, and file-level ownership to answer "What did you contribute?" honestly.

9) Top 10 likely interview questions

- 1. What happens end-to-end when user clicks Analyze Scan?
- 2. Why use three XAI methods instead of one?
- 3. Why is predictor loaded lazily and cached?
- 4. What is stored in the Scan table and why?
- 5. Why choose Django + React for this project?
- 6. What are the biggest security gaps?
- 7. What are the major scalability bottlenecks?
- 8. How does PDF export work in frontend?
- 9. How do list and detail serializers differ?
- 10. If productionizing, what would you change first?

Important File References (quick revision)

- /home/runner/work/Major_project/Major_project/README.md
- /home/runner/work/Major_project/Major_project/backend/api/views.py
- /home/runner/work/Major_project/Major_project/backend/api/inference.py
- /home/runner/work/Major_project/Major_project/backend/api/models.py
- /home/runner/work/Major_project/Major_project/backend/api/serializers.py
- /home/runner/work/Major_project/Major_project/Brain_Tumor_AI/predict.py
- /home/runner/work/Major_project/Major_project/Brain_Tumor_AI/utils/gradcam.py
- /home/runner/work/Major_project/Major_project/Brain_Tumor_AI/utils/shap_explainer.py
- /home/runner/work/Major_project/Major_project/Brain_Tumor_AI/utils/integrated_gradients.py
- /home/runner/work/Major_project/Major_project/frontend/src/pages/Dashboard.jsx
- /home/runner/work/Major_project/Major_project/frontend/src/pages/History.jsx
- /home/runner/work/Major_project/Major_project/frontend/src/pages/ScanDetail.jsx
- /home/runner/work/Major_project/Major_project/frontend/src/api/client.js

Note: This PDF is a compact interview-ready report generated from repository inspection. If you want, I can generate a full expanded PDF with all 30 sections in depth.